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PAPER OF PT1-ACC-24CAI0104-B2024-SEM2 EXAM

Q1. ACC-24CAI1101-MCQ-1MARKS-PT1-B2024-AI

Find the output of the code

```
#include <iostream>
using namespace std;

void modifyArray(int arr[], int size) {
  for(int i = 0; i < size; i++) {
    arr[i] += 5;
  }
}

int main() {
  int arr[3] = {1, 2, 3};
  modifyArray(arr, 3);
  for(int i = 0; i < 3; i++) {
    cout << arr[i] << " ";
  }
  return 0;
}
```

ANSWER:

- ○ 1 2 3
- ◉ 6 7 8
- ○ 1 7 3
- ○ Compilation Error

Ans : 6 7 8

Q2. ACC-24CAI1101-MCQ-1MARKS-PT1-B2024-AI

Identify the error in this code:

```
int x = 5;
int *ptr;
*ptr = &x;
cout << *ptr;
```

ANSWER:

- ○ Syntax error
- ◉ Incorrect assignment to pointer
- ○ No error
- ○ Dereferencing an uninitialized pointer

Ans : Incorrect assignment to pointer

Q3. ACC-24CAI1101-MCQ-1MARKS-PT1-B2024-AI

What will be the output of given function call?

```
int add(int x, int y = 5) {
return x + y;
}
int main(){
cout << add(3);
return 0;
}
```

ANSWER:

- ◉ 8
- ○ 3
- ○ 5
- ○ Syntax error

Ans : 8

Q4. ACC-24CAI1101-MCQ-1MARKS-PT1-B2024-AI

Find the output of the following program.

```
int x = 10;
do {
cout << x;
x--; }
while (x > 10);
```

ANSWER:

- (●) Prints 10 once
- ( ) Prints 10 and then 9
- ( ) Prints numbers from 10 to 1
- ( ) No Output

Ans : Prints 10 once

Q5. ACC-24CAI0104-MCQ-1MARKS-PT1-B2024-AI

Identify the error in the given code.

```
int x = 10;
int y = ++x + 2;
cout << x << " " << y;
```

ANSWER:

- ( ) Incorrect use of increment operator
- (●) No error
- ( ) Syntax error in cout statement
- ( ) Syntax error in cout statement

Ans : No error

Q6. ACC-24CAI0104-CODING-5MARKS-PT1-B2024-AI

There are two exits in a bus with 100 seats:

- First exit is located beside seat number 1
- Second exit is located beside seat number 100

Seats are arranged in a straight line from 1 to 100 with equal spacing between any 2 adjacent seats.

A passenger prefers to choose the nearest exit while leaving the bus.

Determine the exit taken by a passenger sitting on seat X.

Input Format

- The first line of input will contain a single integer T
- *T*, denoting the number of test cases.
- Each test case consists a single integer X
- *X*, denoting the seat number.

Output Format

For each test case, output Left if the passenger chooses the exit beside seat 1, Right otherwise.

Constraints

- $1 \leq T \leq 100$
- $1 \leq X \leq 100$

Sample 1:

Input

6

1

50

100

30

51

73

Output

Left

```
Left

Right

Left

Right

Right
```

ANSWER:

Test case 1 :

Input:
3
10
50
59
Output:
Left
Left
Right

Test case 2 :

Input:
6
3
17
98
51
65
40
Output:
Left
Left
Right
Right
Right
Left

Test case 3 :

Input:
4
99
100
45
23
Output:
Right
Right
Left
Left

Q7. ACC-24CAI0104-CODING-5MARKS-PT1-B2024-AI

Write a program using function that takes an integer **n** as input, representing the number of test cases. For each test case, the program should:

1. Accept two natural numbers **n1** and **n2** such that **n1 < n2**.
2. Compute and print the sum of all even numbers in the inclusive range **[n1, n2]**.

Input Format:

- The first line contains an integer **n** (number of test cases).
- Each of the next **n** lines contains two space-separated natural numbers **n1** and **n2**, where **n1 < n2**.

Output Format:

- For each test case, output a single line containing the sum of even numbers in the range **[n1, n2]**.

Constraints:

- $1 \leq n1 < n2$
- n1, n2 are natural numbers.

Sample Input

2

2 10

5 15

Sample Output

30

40

ANSWER:

Test case 1 :

Input:
2
5 10
10 20
Output:
24
90

Test case 2 :

Input:
3
1 100
50 100
45 50
Output:

2550
1950
144

Q8. ACC-24CAI0104-CODING-5MARKS-PT1-B2024-AI

Move All Zeros to End

You are given **T** test cases. For each test case, you are given an array of **N** integers. Your task is to move all zeros in the array to the end while maintaining the relative order of the non-zero elements. The operation should be performed **in-place**, without using extra space.

Input Format:

- The first line contains an integer **T** — the number of test cases.
- For each test case:
- The first line contains an integer **N** — the size of the array.
- The second line contains **N** space-separated integers representing the array elements.

Output Format:

- For each test case, print the modified array where all zeros are moved to the end while keeping the order of non-zero elements unchanged.

Constraints:

- 1≤T≤100
- 1≤N≤100
- -10^9≤A[i]≤10^9
- The sum of **N** across all test cases does not exceed **10^6**

Example 1:

Input:

2

7

0 1 9 8 4 0 2

5

0 0 0 3 5

Output:

1 9 8 4 2 0 0

```
3 5 0 0 0
```

ANSWER:

Test case 1 :

Input:
3
2
0 0
3
3 5 8
5
0 0 0 8 0
Output:
0 0
3 5 8
8 0 0 0 0

Test case 2 :

Input:
2
2
0 2
3
3 0 5
Output:
2 0
3 5 0